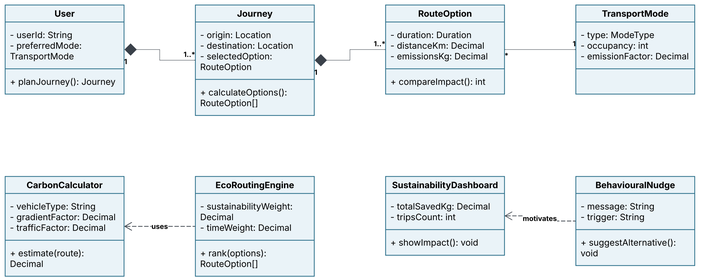
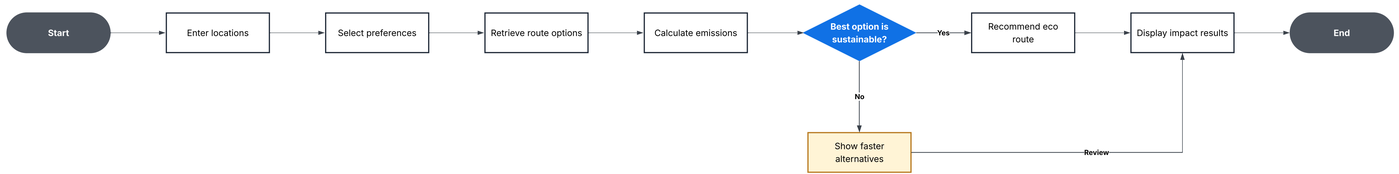

User
- userId: String
- preferredMode: TransportMode
+ planJourney(): Journey
Journey
- origin: Location
- destination: Location
- selectedOption: RouteOption
+ calculateOptions(): RouteOption[]
RouteOption
- duration: Duration
- distanceKm: Decimal
- emissionsKg: Decimal
+ compareImpact(): int
TransportMode
- type: ModeType
- occupancy: int
- emissionFactor: Decimal
1
1..*
1..*
1
*
1
CarbonCalculator
- vehicleType: String
- gradientFactor: Decimal
- trafficFactor: Decimal
+ estimate(route): Decimal
EcoRoutingEngine
- sustainabilityWeight: Decimal
- timeWeight: Decimal
+ rank(options): RouteOption[]
SustainabilityDashboard
- totalSavedKg: Decimal
- tripsCount: int
+ showImpact(): void
BehaviouralNudge
- message: String
- trigger: String
+ suggestAlternative(): void
uses
motivates


Start
Enter locations
Select preferences
Retrieve route options
Calculate emissions
Best option is sustainable?
Yes
No
Recommend eco route
Display impact results
End
Show faster alternatives
Review